Frames and matrix designs

Counting real Hadamard matrices

Difficulty: extreme **Importance:** interesting to the community

Status: Open

Topic: Enumeration of orthogonal sign matrices

Rating rationale: Known general upper bounds still have a quadratic exponent, while the conjecture asks for an exponent of order $n \log n$. The question concerns the abundance of exact flat orthogonal transforms, with connections to structured matrix constructions and elimination.

Statement

For each positive integer n, let

$$H(n) = \#\{A \in \{-1, 1\}^{n\times n} : AA^{\mathsf{T}} = nI_n\}.$$

Conjecture. There is an absolute constant $C > 0$ such that

$$H(n) \le 2^{Cn \log_2 n}$$

for every positive integer n divisible by four.

The count is of individual matrices with their row and column labels. Matrices related by signed permutations are not identified. The statement does not require that a Hadamard matrix exist at every such order.

Known bounds and numerical significance

Ferber, Jain and Zhao prove that some absolute $c_H > 0$ gives $H(n) \le 2^{(1-c_H)n^2/2}$ for every sufficiently large multiple of four. Whenever $H(n) > 0$, distinct row permutations of one Hadamard matrix give $H(n) \ge n!$. Thus the conjectured exponent has the smallest possible order along orders admitting such matrices.

Dividing a Hadamard matrix by $\sqrt{n}$ produces an orthogonal transformation whose entries all have the same magnitude. The enumeration asks how many exact sign designs can underlie these transforms. Peca-Medlin's work connects Hadamard enumeration for butterfly constructions to structured orthogonal matrices and Gaussian elimination. This is distinct from the Hadamard existence conjecture and from bounds on elimination growth for a given matrix.

References and status check

- A. Ferber, V. Jain and Y. Zhao, *On the number of Hadamard matrices via anti-concentration*, Combinatorics, Probability and Computing 31 (2022), 455–477. DOI; published PDF. Conjecture 1.3 on p.456 is the displayed target; Theorem 1.2 gives the upper bound. The arXiv record also supplies the earlier manuscript, where the conjecture has different numbering.
- BIRS workshop 24w5204 report (2024), Conjecture 28, restates the enumeration target.
- J. Peca-Medlin, *Complete pivoting growth of butterfly matrices and butterfly Hadamard matrices* (2026). DOI, §3 before Proposition 3.1, discusses the general count as conjectural and enumerates specific butterfly constructions.

On 2026-09-11, checked the published statement and later restatements, and searched the exact title, the authors' names, Hadamard enumeration, upper bounds, and 2025–2026 proof/counterexample combinations. No full resolution was located. The butterfly counts apply to restricted families and do not settle this all-matrix upper bound. This is a bounded literature check.